hw3-circuits-and-logic: Sample Solutions

CSE20F26

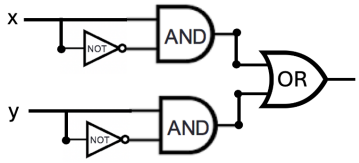
In this assignment, you will consider how circuits and logic can be used to represent mathematical claims. You will use propositional operators to express and evaluate these claims. You will analyze statements and determine if they are true or false using valid proof strategies.

Relevant class material: Week 3, 4, and 5.

Assigned questions

1. Circuits.

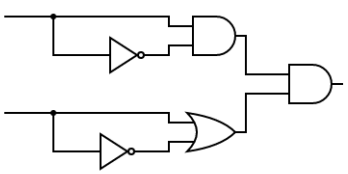
   (a) Consider the circuit below with inputs x and y. Can one of the AND gates be exchanged with the OR gate without changing the input-output table of the circuit? If so, write out the input-output table that results, and briefly explain why this exchange works. If not, explain why not with reference to the definitions of the logic gates.



Sample solution:

Input-output table of the original circuit:

Input-output table				
x	y	$x \wedge \neg x$	$y \wedge \neg y$	$output = (x \wedge \neg x) \vee (y \wedge \neg y)$
1	1	0	0	0
1	0	0	0	0
0	1	0	0	0
0	0	0	0	0



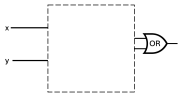
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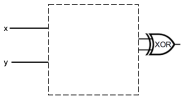
Switch: We can switch one of the AND gate with the OR gate. For example, we can switch the AND gate connected with y with the OR gate (as shown in the circuit above), and the new table would look like:

Input-output table				
x	y	$x \wedge \neg x$	$y \vee \neg y$	$output = (x \wedge \neg x) \wedge (y \vee \neg y)$
1	1	0	1	0
1	0	0	1	0
0	1	0	1	0
0	0	0	1	0

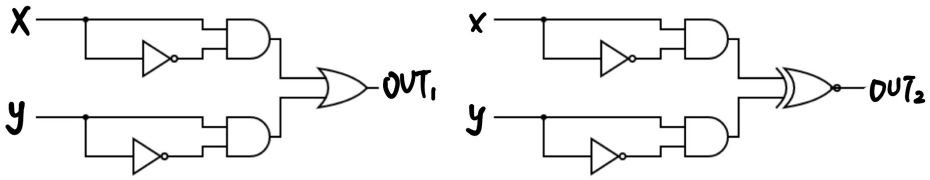
Explanation: Even though $y \vee \neg y$ will always output 1, we still have the $x \wedge \neg x$ which will always output 0. Since we moved one AND gate to the end, the input 0 to the AND gate will make sure the final output will still be constantly 0.

(b) Is there a way to fill in the blank portion of the two logic circuits below *with the same gates connected in the same way* so that the resulting circuits have the same input-output value *even though* one uses an OR gate at the end and the other uses an XOR gate? If so, design the circuit that would be used, write out the input-output table that results, and briefly explain why your design works. If not, explain why not with reference to the definitions of the logic gates.





Sample solution:
Explanation: From 2(a), we learned that $x \wedge \neg x$ will always produce 0. If we combine it with the fact that $0 \vee 0 = 0 = 0 \oplus 0$, we can make a circuit where the two inputs to the final gate are both 0:



Input-output table:

Input-output table					
x	y	$x \wedge \neg x$	$y \wedge \neg y$	$OUT_1 = (x \wedge \neg x) \vee (y \wedge \neg y)$	$OUT_2 = (x \wedge \neg x) \oplus (y \wedge \neg y)$
1	1	0	0	0	0
1	0	0	0	0	0
0	1	0	0	0	0
0	0	0	0	0	0

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2. Compound propositions. The set of strings of length 4 whose characters are 0s or 1s is the result of four successive set-wise concatenations: $\{0,1\} \circ \{0,1\} \circ \{0,1\} \circ \{0,1\}$. Let's call this set X_4. Consider the function $f : X_4 \to X_4$ defined by

$$f(x) = \begin{cases} y & \text{when } (x)_{2,4} < 15 \text{ and } (y)_{2,4} = (x)_{2,4} + 1 \\ 1111 & \text{when } x = 1111 \end{cases}$$

for each $x \in X_4$. In other words, we can describe the function as: f takes a string, interprets it as the binary fixed-width 4 expansion of an integer, and then adds 1 to that integer (unless x is already representing the greatest integer that can be represented in binary fixed-width 4) and outputs the binary fixed-width 4 expansion of the result.

(a) Fill in the blanks in the following input-output definition table with four inputs x_3, x_2, x_1, x_0 and four outputs y_3, y_2, y_1, y_0 so that $f(x_3x_2x_1x_0) = y_3y_2y_1y_0$. Remember to explain all your answers.

x_3	x_2	x_1	x_0	y_3	y_2	y_1	y_0
1	1	1	1	1	1	BLANK1	1
1	1	1	0	1	1	1	1
1	1	0	1	1	BLANK2	1	0
1	1	0	0	1	1	0	1
1	0	1	1	1	1	0	0
1	0	1	0	1	0	1	1
1	0	0	1	1	0	1	0
1	0	0	0	BLANK3	0	0	1
0	1	1	1	1	0	0	0
0	1	1	0	0	1	1	BLANK4
0	1	0	1	0	1	1	BLANK5
0	1	0	0	0	1	0	1
0	0	1	1	0	1	0	0
0	0	1	0	0	0	1	1
0	0	0	1	0	0	1	0
0	0	0	0	BLANK6	0	0	1

Solution:

$f(1111) = 1111$, so BLANK1 $= 1$.

$f(1101)$ is the bit string representing $(1101)_{2,4} + 1 = (1110)_{2,4}$, so BLANK2 $= 1$.

$f(1000)$ is the bit string representing $(1000)_{2,4} + 1 = (1001)_{2,4}$, so BLANK3 $= 1$.

$f(0110)$ is the bit string representing $(0110)_{2,4} + 1 = (0111)_{2,4}$, so BLANK4 $= 1$.

$f(0101)$ is the bit string representing $(0101)_{2,4} + 1 = (0110)_{2,4}$, so BLANK5 $= 0$.

$f(0000)$ is the bit string representing $(0000)_{2,4} + 1 = (0001)_{2,4}$, so BLANK6 $= 0$.

Final table:

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x_3	x_2	x_1	x_0	y_3	y_2	y_1	y_0
1	1	1	1	1	1	1	1
1	1	1	0	1	1	1	1
1	1	0	1	1	1	1	0
1	1	0	0	1	1	0	1
1	0	1	1	1	1	0	0
1	0	1	0	1	0	1	1
1	0	0	1	1	0	1	0
1	0	0	0	1	0	0	1
0	1	1	1	1	0	0	0
0	1	1	0	0	1	1	1
0	1	0	1	0	1	1	0
0	1	0	0	0	1	0	1
0	0	1	1	0	1	0	0
0	0	1	0	0	0	1	1
0	0	0	1	0	0	1	0
0	0	0	0	0	0	0	1

(b) Construct an expression (as a compound proposition) for each of y_0, y_1, y_2, and y_3 in terms of the inputs x_3, x_2, x_1, x_0. Justify your expression by referring to the definition of the logic gates XOR, AND, OR, NOT and the definition of the function f. Hint: our work on the half-adder might be helpful.

Solution: The function f takes a length-4 bit string $x = x_3x_2x_1x_0$, outputs the fixed-width 4 binary representation of $(x)_{2,4} + 1$ (unless $x = 1111$). The bits y_0, y_1, y_2, y_3 are what x_0, x_1, x_2, x_3 become after incrementing by 1, respectively. We observe that for any bit $b \in \{0, 1\}$, XOR-ing b with 1 essentially flips the bit, i.e. $b \oplus 1 = \neg b$, and XOR-ing b with 0 does not change the bit, i.e. $b \oplus 0 = b$. Moreover, if $x = 1111$, i.e. (when the proposition $(x_0 \wedge x_1 \wedge x_2 \wedge x_3)$ evaluates to TRUE), we have $y_0 = y_1 = y_2 = y_3 = 1$ as $f(1111) = 1111$.

i. Consider the last bit x_0. If $x_0 = 0$, then incrementing $(x)_{2,4}$ by 1 will cause x_0 to flip into 1, and vice versa. Therefore,

$$y_0 = \neg x_0 \vee (x_0 \wedge x_1 \wedge x_2 \wedge x_3),$$

where $(x_0 \wedge x_1 \wedge x_2 \wedge x_3)$ handles the edge case.

ii. When incrementing by 1, x_1 flips when $x_0 = 1$, and stays the same otherwise. Therefore, we can write

$$y_1 = (x_1 \oplus x_0) \vee (x_0 \wedge x_1 \wedge x_2 \wedge x_3)$$

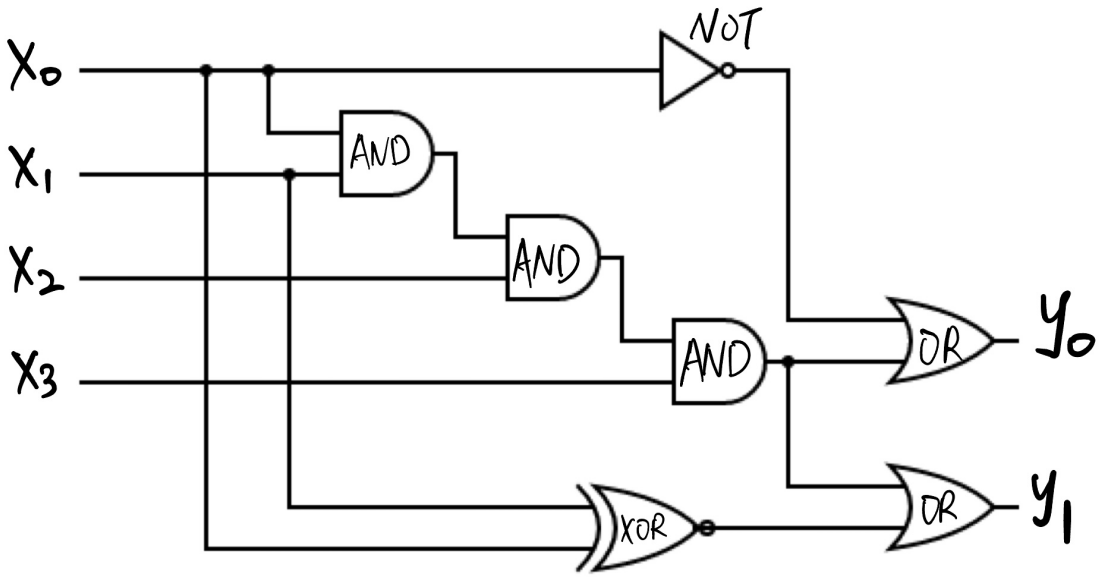
iii. When incrementing by 1, x_2 flips when the right-most bits x_1, x_0 are both 1, and stays the same otherwise. Therefore, we can write

$$y_2 = (x_2 \oplus (x_0 \wedge x_1)) \vee (x_0 \wedge x_1 \wedge x_2 \wedge x_3)$$

iv. Similarly, when incrementing by 1, x_3 flips when the right-most bits x_2, x_1, x_0 are all 1, and stays the same otherwise. Therefore, we can write

$$y_3 = (x_3 \oplus (x_0 \wedge x_1 \wedge x_2)) \vee (x_0 \wedge x_1 \wedge x_2 \wedge x_3)$$

(c) Draw a combinatorial circuit corresponding to two (or more) of these (output) compound propositions. Remember that the symbols for the inputs will be on the left-hand-side and the symbol for the outputs y_0 and/or y_1 and/or y_2 and/or y_3 will be on the right-hand side. Use gates (draw the appropriate shapes and add labels for clarity) and wires to connect the inputs appropriately to give the output. **Solution:**



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3. Logical Equivalence. Imagine a friend suggests the following argument to you: "The compound proposition

$$(x \vee y) \wedge z$$

is logically equivalent to

$$x \vee (y \wedge z)$$

because I can transform one to the other using the following sequence of logical equivalences:

$$(x \vee y) \wedge z \equiv (x \vee (y \wedge y)) \wedge z \equiv x \vee ((y \vee y) \wedge z) \equiv x \vee (y \wedge z)$$

because y is logically equivalent to both $y \wedge y$ and to $y \vee y$".

(a) Prove to your friend that they made a mistake by giving a truth assignment to the propositional variables x, y, z so that the two compound propositions $(x \vee y) \wedge z$ and $x \vee (y \wedge z)$ have different truth values. Justify your choice by evaluating these compound propositions using the definitions of the logical connectives and include enough intermediate steps so that a student in CSE 20 who may be struggling with the material can still follow along with your reasoning.

Sample Solution:

Idea: By observing that the second proposition is always true when x is true, we will be trying to find an assignment to y and z to make the first proposition false when x is true. Then we can spot that $(x \vee y) \wedge z$ is false when z is false, which will result in the first proposition being false.

Assignment: $x = \mathrm{T}$, $y = \mathrm{T}$, $z = \mathrm{F}$.

The row of the input-output table corresponding to this assignment is:

Input-output table						
x	y	z	$x \vee y$	$y \wedge z$	$(x \vee y) \wedge z$	$x \vee (y \wedge z)$
T	T	F	T	F	F	T

(b) Help your friend find the problem in their argument by pointing out which step(s) were incorrect and why.

Sample Solution:

If we evaluate every proposition in the friend's sequence of the logical equivalences, we can see that, for a specific truth assignment, the consecutive propositions in the argument do not have the same truth value:

Input-output table						
x	y	z	$(x \vee y) \wedge z$	$(x \vee (y \wedge y)) \wedge z$	$x \vee ((y \vee y) \wedge z)$	$x \vee (y \wedge z)$
T	F	T	F	F	T	T

Hence, we can conclude that the problem in the argument must be the step where the friend claims: $(x \vee (y \wedge y)) \wedge z \equiv x \vee ((y \vee y) \wedge z)$.

(c) Give **three** different compound propositions that are actually logically equivalent to (and not the same as)

$$(x \vee y) \wedge z$$

Justify each one of these logical equivalences either by applying a sequence of logical equivalences or using a truth table. Notice that you can use other logical operators (e.g. $\neg, \vee, \wedge, \oplus, \rightarrow, \leftrightarrow$) when constructing your compound propositions.

Bonus; not for credit (do not hand in): How would you translate each of the equivalent compound propositions in English? Does doing so help illustrate why they are equivalent?

Sample solution:

Easy ones: using $a \equiv a \vee a \equiv a \wedge a \equiv \neg\neg a$, we can change the proposition to:

$$\begin{aligned}(x \vee y) \wedge z &\equiv ((x \vee x) \vee y) \wedge z \\ &\equiv ((x \vee x) \vee y) \wedge (z \wedge z) \\ &\equiv (x \vee y) \wedge (\neg\neg z) \qquad \text{etc.}\end{aligned}$$

More advanced ones: we can apply DeMorgan's Laws:

$$\begin{aligned}(x \vee y) \wedge z &= \neg(\neg(x \vee y) \vee \neg z) \\ &= \neg((\neg x \wedge \neg y) \vee \neg z)\end{aligned}$$

4. Evaluating predicates. Consider the following predicates, each of which has as its domain the set of all bitstrings whose leftmost bit is 1

   $E(x)$ is T exactly when $(x)_2$ is even, and is F otherwise

   $L(x)$ is T exactly when $(x)_2 < 15$, and is F otherwise

   $M(x)$ is T exactly when $(x)_2 > 2$ and is F otherwise.

Sample response that can be used as reference for the detail expected in your answer: To prove that the statement

$$\forall x\ L(x)$$

is false, we can use the counterexample $x = 1111$, which is a bitstring whose leftmost bit is 1 (so is in the domain). Applying the definition of $L(x)$, since $(1111)_2 = 1 \cdot 2^3 + 1 \cdot 2^2 + 1 \cdot 2^1 + 1 \cdot 2^0 = 8 + 4 + 2 + 1 = 15$ which is not (strictly) less than 15, we have that $L(1111) = F$ and so the universal statement is false.

(a) Use a counterexample to prove that the statement

$$\forall x(\ L(x) \to E(x)\ )$$

is false. Your proof needs to include references to the relevant definitions.

Sample solution:

Idea: The counterexample needs to make $L(x)$ True (i.e. $(x)_2 < 15$), and $E(x)$ False (i.e. $(x)_2$ is not even).

My counterexample is $x = 11$, which is a bitstring whose leftmost bit is 1 (so is in the domain). Applying the definition of $L(x)$, since $(11)_2 = 1 \cdot 2^1 + 1 \cdot 2^0 = 3$ which is strictly less than 15, we have that $L(11) = T$. Applying the definition of $E(x)$, since $(11)_2 = 3$ is odd, we have that $E(11) = F$. Thus, $L(11) \to E(11) = T \to F = F$, and the statement $\forall x(\ L(x) \to E(x)\ )$ is false.

(b) Use a witness to prove that the statement

$$\exists x(\ L(x) \wedge M(x)\ )$$

is true. Your proof needs to include references to the relevant definitions.

Sample solution:

Idea: the witness needs to make $L(x)$ True (i.e. $(x)_2 < 15$), and $M(x)$ True (i.e. $(x)_2 > 2$).

My witness is $x = 100$, which is a bitstring whose leftmost bit is 1 (so is in the domain). Applying the definition of $L(x)$, since $(100)_2 = 1 \cdot 2^2 = 4$, which is strictly less than 15, we have that $L(100) = T$. Applying the definition of $M(x)$, since $(100)_2 = 4$ is strictly greater than 2, we have that $M(100) = T$. Therefore, $L(100) \wedge M(100) = T$, and the statement $\exists x(\ L(x) \wedge M(x)\ )$ is true.

(c) Translate each of the statements in the previous two parts to English.

Sample solutions: The statement in part (a) says, "For every bitstring x that can be interpreted as the binary expansion of an integer, if that integer is strictly less than 15, then x is even."

The statement in part (b) says, "There is a bitstring that is the binary expansion of an integer that is strictly less than 15 and strictly greater than 2."

5. Set properties. Let $W = \mathcal{P}(\{1, 2, 3, 4, 5\})$.

Sample response that can be used as reference for the detail expected in your answer for parts (a) and (b) below:

To give an example element in the set $\{X \in W \mid 1 \in X\} \cap \{X \in W \mid 2 \in X\}$, consider $\{1, 2\}$. To prove that this is in the set, by definition of intersection, we need to show that $\{1, 2\} \in \{X \in W \mid 1 \in X\}$ and that $\{1, 2\} \in \{X \in W \mid 2 \in X\}$.

- By set builder notation, elements in $\{X \in W \mid 1 \in X\}$ have to be elements of W which have 1 as an element. By definition of power set, elements of W are subsets of $\{1, 2, 3, 4, 5\}$. Since each element in $\{1, 2\}$ is an element of $\{1, 2, 3, 4, 5\}$, $\{1, 2\}$ is a subset of $\{1, 2, 3, 4, 5\}$ and hence is an element of W. Also, by roster method, $1 \in \{1, 2\}$. Thus, $\{1, 2\}$ satisfies the conditions for membership in $\{X \in W \mid 1 \in X\}$.
- Similarly, by set builder notation, elements in $\{X \in W \mid 2 \in X\}$ have to be elements of W which have 2 as an element. By definition of power set, elements of W are subsets of $\{1, 2, 3, 4, 5\}$. Since each element in $\{1, 2\}$ is an element of $\{1, 2, 3, 4, 5\}$, $\{1, 2\}$ is a subset of $\{1, 2, 3, 4, 5\}$ and hence is an element of W. Also, by roster method, $2 \in \{1, 2\}$. Thus, $\{1, 2\}$ satisfies the conditions for membership in $\{X \in W \mid 2 \in X\}$.

(a) Give two (different) example elements in

$$W \times \mathcal{P}(W)$$

Justify your examples by explanations that include references to the relevant definitions.

Sample solution: Two examples are

$$(\{1, 2\}, \{\{1, 3, 4\}, \{2, 3\}\})$$

and

$$(\emptyset, \{\emptyset\})$$

which are each elements in $W \times \mathcal{P}(W)$.

Each of these is an ordered pair (so it's a candidate for being in the Cartesian product of two sets), and we need to describe how the first component of the ordered pair is in W and how the second component of the ordered pair is in $\mathcal{P}(W)$.

Since W is the power set of $\{1, 2, 3, 4, 5\}$, every element in W is a subset of $\{1, 2, 3, 4, 5\}$. Thus, $\{1, 2\} \in W$ and $\emptyset \in W$ (as the empty set is a subset of every set). Since $\mathcal{P}(W)$ is the power set of W, every element in $\mathcal{P}(W)$ is a subset of elements in W, and so must be a set of subsets of $\{1, 2, 3, 4, 5\}$. Thus, $\{\{1, 3, 4\}, \{2, 3\}\} \in \mathcal{P}(W)$ because $\{1, 3, 4\} \in W$ and $\{2, 3\} \in W$. Likewise, $\{\emptyset\} \in \mathcal{P}(W)$ because $\emptyset \in W$.

(b) Give one example element in

$$\{X \in W \mid (1 \in X) \wedge (X \cap \{3,4\} = \emptyset)\}$$

Justify your example by explanations that include references to the relevant definitions.

Sample solution: One example is $\{1,2\}$. To prove that this is in the set, by the set builder notation and the definition of conjunction, we need to show that $\{1,2\} \in W$, $1 \in \{1,2\}$, and $\{1,2\} \cap \{3,4\} = \emptyset$.

- By definition of power set, elements of W are subsets of $\{1,2,3,4,5\}$. Since each element in $\{1,2\}$ is an element of $\{1,2,3,4,5\}$, $\{1,2\}$ is a subset of $\{1,2,3,4,5\}$ and hence is an element of W.
- Since 1 is listed as one of the elements of the set in the roster notation definition $\{1,2\}$, we know $1 \in \{1,2\}$.
- By definition of set intersection, $\{1,2\} \cap \{3,4\} = \{x | x \in \{1,2\} \wedge x \in \{3,4\}\}$. If we consider the elements x where $x \in \{1,2\}$, the only elements that evaluate to true are 1 and 2. However, neither of these are elements of $\{3,4\}$, which means there are no elements where $x \in \{1,2\}$ and $x \in \{3,4\}$ are both satisfied. Hence, we get $\{1,2\} \cap \{3,4\} = \emptyset$ as desired.

(c) Consider the following statement and attempted proof:

$$\forall A \in W \, \forall B \in W \;\; (\; ((A \cap B) \subseteq A) \rightarrow (A \subseteq B) \;)$$

(1) Towards a universal generalization argument, **choose arbitrary** $A \in W, B \in W$.
(2) We need **to show** $((A \cap B) \subseteq A) \rightarrow (A \subseteq B)$.
(3) Towards a proof of the conditional by the contrapositive, **assume** $A \subseteq B$, and we need **to show** that $(A \cap B) \subseteq A$.
(4) By definition of subset inclusion, this means we need **to show** $\forall x \; (x \in A \cap B \rightarrow x \in A \;)$.
(5) Towards a universal generalization, **choose arbitrary** x; we need **to show** that
$x \in A \cap B \rightarrow x \in A$.
(6) Towards a direct proof, **assume** $x \in A \cap B$, and we need **to show** $x \in A$.
(7) By definition of set intersection and set builder notation, we have that $x \in A \wedge x \in B$.
(8) By the definition of conjunction, $x \in A$, which is what we needed to show. QED

Demonstrate that this attempted proof is invalid by providing and justifying a **counterexample** (disproving the statement). Then, explain why this attempted proof is invalid by identifying in which step a definition or proof strategy is used incorrectly, and describing how the definition or proof strategy was misused.

Sample Solution:

Counterexample:

One counterexample is $A = \{1, 2\}$ and $B = \{1\}$. As proven above, they are both subset of $\{1, 2, 3, 4, 5\}$ and hence in W. By definition of set intersection, we can show $A \cap B = \{1\}$, which is a subset of A. Hence, $(A \cap B) \subseteq A$ is true.

However, there exists $2 \in A$, but $2 \notin B$, which means A is not a subset of B, and $A \subseteq B$ evaluates to False.

Hence, $(A \cap B) \subseteq A) \rightarrow (A \subseteq B)$ is False for our counterexample $A = \{1, 2\}$ and $B = \{1\}$, and thus the universal statement is False.

Misuse in the proof:

In (3), the "contrapositive" was misused. The definition of contrapositive is "to prove that the implication $p \rightarrow q$ is true, we can assume q is **false** and use that assumption to show p is also **false**" . The step (3) of the attempted proof was assuming q is **True**, and was trying to prove p is also **True**. The correct use of contrapositive in this case will be: "Towards a proof of the conditional by the contrapositive, assume A is **not** a subset of B, and we need to show that $(A \cap B)$ is **not** a subset of A."